

Route G-R2: Two Events, Two Durable Records

A finite relational protocol, not a density–time law

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Abstract

A cyclic clock reading is not itself a saved event. We construct one initial record and one terminal readout, including their quantum backreaction, inside an exact finite history constraint. Orthogonal memory sectors retain the outcomes and two fixed protocol stamps. A one-bit epoch label separates repeated clock phases. The construction is conditional on an ordered protocol and a blank-memory boundary; it does not derive an arrow of time, apparatus energetics, or an E_d law. In an exact three-level example the terminal probability is 0 without the initial record, $2/9$ with it, and $4/9$ for a different instrument having the same initial measurement effects.

1 Declared system, clock and recording gates

G-R1 supplied a noninteracting finite Page–Wootters constraint and single-reading statistics. Recording requires an additional coupling. Here we retain its three-level free conditional propagator,

$$H_S = \epsilon \operatorname{diag}(0, 1, 2), \quad U = \operatorname{diag}(1, z, z^2), \quad z = e^{-2\pi i/3}, \quad U^3 = \mathbb{I}, \quad (1)$$

but *replace* its constraint by the explicit protocol constraint below. No claim is made that the old $H_C + H_S - E_*$ implements these gates. All phases are clock labels, not spatial rotations.

The work space is $W = S \otimes A \otimes F$, of dimension 27. Each memory has orthogonal states $|\emptyset\rangle, |0\rangle, |1\rangle$. Let S_a exchange $|\emptyset\rangle$ and $|a\rangle$, leaving the third state fixed. With $|+\rangle = (|0\rangle + |1\rangle + |2\rangle)/\sqrt{3}$, set

$$P_0 = |0\rangle\langle 0|, \quad P_1 = \mathbb{I} - P_0, \quad Q_+ = |+\rangle\langle +|, \quad Q_- = \mathbb{I} - Q_+, \quad (2)$$

$$W_i = \sum_{a=0,1} P_a \otimes S_a \otimes \mathbb{I}, \quad W_f = \sum_{b=+,-} Q_b \otimes \mathbb{I} \otimes S_b. \quad (3)$$

For F , S_+ denotes S_0 , and S_- denotes S_1 . Orthogonality and completeness of P_a, Q_b prove $W_i^\dagger W_i = W_f^\dagger W_f = \mathbb{I}$. These are unitary premeasurements; a later pointer read gives the outcome. For a blank memory,

$$W_i |\psi\rangle |\emptyset\rangle_A |\emptyset\rangle_F = \sum_a P_a |\psi\rangle |a\rangle_A |\emptyset\rangle_F. \quad (4)$$

This is standard instrument physics, not a new measurement law. Explicit memories in conditional-time models are discussed in [1]; our finite boundary/epoch choice is declared here.

2 An exact finite stationary history, with an explicit boundary

Take the orthonormal G-R1 clock grid $|\varphi_k\rangle = 3^{-1/2} \sum_{n=0}^2 e^{2\pi i n k/3} |n\rangle$. Add an epoch bit R , and use

$$|q\rangle = |\varphi_{q \bmod 3}\rangle_C | \lfloor q/3 \rfloor \rangle_R, \quad 0 \leq q \leq 5. \quad (5)$$

This is six distinct subsystem configurations, not six externally calibrated times. The bit is not an unlimited cycle counter. For $\chi_0 = |+\rangle|\emptyset\rangle_A |\emptyset\rangle_F$ define $U_W = U \otimes \mathbb{I}_A \otimes \mathbb{I}_F$ and

$$(V_0, V_1, V_2, V_3, V_4) = (W_i, U_W, W_f U_W, U_W, U_W), \quad \chi_{q+1} = V_q \chi_q. \quad (6)$$

Only one initial gate and one final gate are applied. Two free propagators separate the recorded initial state χ_1 from the terminal measurement in χ_3 ; hence $U_\Delta = U^2$. Recording links are not asserted to have the duration of a free tick.

The memory labels may be read as $|(r_i, k_i), a\rangle_A$ and $|(r_f, k_f), b\rangle_F$, with fixed stamps $(r_i, k_i) = (0, 1)$ and $(r_f, k_f) = (1, 0)$, using the post-gate configuration convention. Because each event has a predeclared stamp, three memory states per event suffice. This is *not* an autonomous recorder that measures an arbitrary unknown clock reading. The gate position supplies the stamp; a general timestamp recorder would require additional states.

Define row maps and a positive operator of dimension 162:

$$A_q = \langle q+1 | \otimes \mathbb{I} - \langle q | \otimes V_q, \quad (7)$$

$$K_{\text{prop}} = \frac{1}{2} \sum_{q=0}^4 A_q^\dagger A_q, \quad K_{\text{in}} = |0\rangle\langle 0| \otimes (\mathbb{I} - |\chi_0\rangle\langle\chi_0|), \quad (8)$$

$$K_{\text{hist}} = K_{\text{prop}} + K_{\text{in}}, \quad |\Omega\rangle = \frac{1}{\sqrt{6}} \sum_{q=0}^5 |q\rangle \chi_q. \quad (9)$$

This is a finite history-state construction in the tradition of [2], with all terms specified rather than a continuum ideal clock. For any $|X\rangle = \sum_q |q\rangle x_q$,

$$\langle X | K_{\text{hist}} | X \rangle = \frac{1}{2} \sum_{q=0}^4 \|x_{q+1} - V_q x_q\|^2 + \|(\mathbb{I} - |\chi_0\rangle\langle\chi_0|)x_0\|^2 \geq 0. \quad (10)$$

Thus $K_{\text{hist}}\Omega = 0$. Conversely, zero energy in (10) implies $x_0 = c\chi_0$ and $x_{q+1} = V_q x_q$, so the kernel is exactly one-dimensional. Conditioning on q gives χ_q and $p(q) = 1/6$.

The ordered links and blank-input boundary are assumptions, not an emergent causal order. Positivity of this protocol constraint is not an apparatus energy budget, and its eigenvalues have no assigned physical energy scale. We have not retained G-R1's separate proof of a noninteracting, fixed-total-local-energy realization.

3 Sequential probabilities and an exact backreaction test

For initial density matrix ρ , instruments with Kraus operators M_a, N_b satisfying $\sum_a M_a^\dagger M_a = \sum_b N_b^\dagger N_b = \mathbb{I}$ give the joint memory statistics

$$p(a, b) = \text{tr} \left[N_b U_\Delta M_a \rho M_a^\dagger U_\Delta^\dagger N_b^\dagger \right]. \quad (11)$$

Indeed, for pure input the completed record state is $\sum_{a,b} N_b U_\Delta M_a |\psi\rangle |a\rangle_A |b\rangle_F$. The squared norm in the orthogonal pointer sector (a, b) yields (11); linearity gives mixed inputs. Completeness gives

$$\sum_b p(a, b) = \text{tr}(M_a \rho M_a^\dagger) = p(a), \quad \sum_{a,b} p(a, b) = 1, \quad p(b | a) = p(a, b)/p(a) \quad (12)$$

when $p(a) > 0$. These are standard sequential Born probabilities [1], not a product of separate unperturbed readings. Here they also follow directly by projecting Ω onto $q = 3$ and the two pointer sectors.

For our projective instrument $M_a = P_a, N_b = Q_b$ and $\rho = |+\rangle\langle+|$,

$$\langle + | U^2 P_0 | + \rangle = 1/3, \quad (13)$$

$$\langle + | U^2 P_1 | + \rangle = (z^2 + z^4)/3 = -1/3. \quad (14)$$

Consequently

$$[p(a, b)]_{\substack{a=0,1 \\ b=+,-}} = \frac{1}{9} \begin{pmatrix} 1 & 2 \\ 1 & 5 \end{pmatrix}, \quad p(a) = (1/3, 2/3), \quad p(+ | a) = (1/3, 1/6). \quad (15)$$

The minus entries are the corresponding branch norms $\|P_a |+\rangle\|^2$ minus the plus entries. Ignoring the first pointer, but actually creating its record, gives

$$\rho \mapsto \sum_a P_a \rho P_a, \quad p(+) = 2/9. \quad (16)$$

By contrast, omitting the initial interaction altogether gives

$$p_{\text{no record}}(+) = |\langle + | U^2 | + \rangle|^2 = |1 + z^2 + z^4|^2/9 = 0. \quad (17)$$

The difference is measurement backreaction, not a changed mass or a slower clock. The rank-two branch retains coherence internally; the initial instrument removes only coherence between the two P_a sectors.

Why the effects alone do not specify the protocol

Take $R_0 = \mathbb{I}$, $R_1 = \text{diag}(1, 1, -1)$ and $M'_a = R_a P_a$. Then $M'_a{}^\dagger M'_a = P_a$: the initial outcome probabilities are unchanged. But the plus amplitude in branch 1 is $(z^2 - z^4)/3$, of squared norm $1/3$. Thus

$$[p'(a, b)] = \frac{1}{9} \begin{pmatrix} 1 & 2 \\ 3 & 3 \end{pmatrix}, \quad p'(+) = 4/9. \quad (18)$$

A controlled R_a on the already written pointer implements this alternative unitarily. It is an audit of instrument dependence, not an activated E_d interaction.

4 Saved records, recurrent readings and the cycle obstruction

Let $\Pi_a^A = \mathbb{I}_S \otimes |a\rangle\langle a|_A \otimes \mathbb{I}_F$ and define Π_b^F similarly, including the blank sector. Every link after writing A commutes with Π_a^A ; every link after writing F commutes with both pointers. Hence

$$\langle \chi_q | \Pi_a^A \Pi_b^F | \chi_q \rangle = p(a, b), \quad q = 3, 4, 5. \quad (19)$$

In fact the entire reduced state of A is unchanged by any subsequent unitary acting only on S, F . Conditioning on a terminal outcome can update our inferred distribution of a but does not overwrite A . Repeated projective pointer reads are nondemolition since $\Pi_a \Pi_{a'} = \delta_{aa'} \Pi_a$. The nonselective read map $\rho_W \mapsto \sum_{a,b} \Pi_a^A \Pi_b^F \rho_W \Pi_a^A \Pi_b^F$ preserves all pointer probabilities.

This does not assert that the joint memory density matrix has already lost every off-diagonal entry: premeasurement can retain coherences. Nor is an arbitrary quantum state cloned. The records are distinguishable pointer sectors with a repeatable classical readout.

With $\Pi_{\text{filled}} = \mathbb{I} - |\emptyset\rangle\langle\emptyset|$, define the two-event counter

$$N_{\text{evt}} = \mathbb{I}_S \otimes \Pi_{\text{filled}}^A \otimes \mathbb{I}_F + \mathbb{I}_S \otimes \mathbb{I}_A \otimes \Pi_{\text{filled}}^F. \quad (20)$$

Each slice is an exact eigenstate, $N_{\text{evt}} \chi_q = n_q \chi_q$:

configuration q	0	1	2	3	4	5
clock reading k	0	1	2	0	1	2
epoch r	0	0	0	1	1	1
stored-event count n_q	0	1	1	2	2	2

Only the two declared recording events are counted, not every phase tick. This is not a universal identification with the source's T or energy information.

Ignoring the epoch and conditioning only on $k = 0$ leaves

$$\rho_{W|k=0} = \frac{1}{2}(|\chi_0\rangle\langle\chi_0| + |\chi_3\rangle\langle\chi_3|), \quad \Pr(F \text{ filled} \mid k = 0) = \frac{1}{2}. \quad (21)$$

Reading $r = 1$ as well, or selecting the completed terminal-memory sector, instead singles out χ_3 at this phase. A cyclic phase without a record or a cycle tag does not identify a unique event.

Suppose instead we forced the entire work state to be periodic after this episode while starting both memories blank. Closure would require

$$W_{\text{cyc}} \chi_0 = \chi_0, \quad W_{\text{cyc}} = W_f U_W^2 W_i. \quad (22)$$

But the left side has both pointers filled, whereas the right side has both blank. They are orthogonal, so

$$\langle \chi_0 | W_{\text{cyc}} \chi_0 \rangle = 0, \quad \|(W_{\text{cyc}} - \mathbb{I}) \chi_0\| = \sqrt{2}. \quad (23)$$

The incompatible requirements are full-state recurrence and a fresh, retained record, not cyclic clocks in general. Resetting, unwriting, a longer recurrence, or an explicit epoch/memory extension changes the requirements. Our open history uses the last option.

5 What this closes, and what it deliberately leaves open

The bounded G-R2 question is answered: specified record interactions convert two scheduled cyclic-clock events into orthogonal, readable memory sectors. One static joint history yields the complete two-event Born statistics, including backreaction and retained records. The physical information added to a recurrent phase is *which episode, which event, and which instrument outcome*, not a fitted function of energy density.

There are three important qualifications.

1. *Finite protocol, not an autonomous universal clock.* The epoch bit, two fixed stamps, ordered gates and input boundary are explicit inputs. No general unknown timestamp recorder, infinite counter or uniquely derived event algebra is claimed. We do not claim the separate epoch register is dimension-minimal; in other protocols memory occupancy can also distinguish stages.
2. *Preservation, not thermodynamic irreversibility.* The records persist under the specified future links and could persist under further isolated-pointer links. Real memory lifetime, environmental errors, redundant records and erasure costs are not supplied. In this model $W_{\text{cyc}}^\dagger$ restores the blank state exactly. The stationary history and its oriented boundary therefore do not derive a thermodynamic arrow.
3. *Constraint consistency, not physical energy completion.* $[W_i, H_S \otimes \mathbb{I}_{AF}] = 0$, but $[W_f, H_S \otimes \mathbb{I}_{AF}] \neq 0$ because $[Q_+, H_S] \neq 0$. Work/clock/apparatus degrees of freedom are needed for a physical energy-conserving realization of that readout. $K_{\text{hist}}\Omega = 0$ does not by itself supply this realization. A microscopic clock speed or gate duration cannot be extracted from its arbitrary overall normalization.

Verification and reproducibility

The executable companion `route_g/code/verify_gr2_two_event_records.py` constructs the gates and the full 162×162 constraint independently of the analytic probabilities. Its 139 checks cover unitarity, projector completeness, sum-of-squares assembly, the unique zero mode, all conditioned slices, both exact probability tables, the no-record test, memory preservation, phase aliasing, failed periodic closure, reversal, mixed inputs and unchanged prior source files. These certify finite algebra, not empirical physics. The JSON report retains individual residuals and source hashes.

Next decision, not executed in G-R2

For G-R3, choose an explicit E_i/E_d object and reference measure, then one clock interaction and a reference-clock comparison. In particular, distinguish a state-dependent interaction from a common energy rescaling or a relabeling of one clock. No slowdown law is selected, fitted or inferred here. The original density–time proposal, emergent spacetime, gravity, and all Route-F promotion gates remain open.

References

- [1] V. Giovannetti, S. Lloyd and L. Maccone, *Quantum Time*, Phys. Rev. D **92**, 045033 (2015), arXiv:1504.04215v3, especially the Measurements section, Eqs. (31)–(46).
- [2] D. Aharonov et al., *Adiabatic Quantum Computation is Equivalent to Standard Quantum Computation*, SIAM J. Comput. **37**, 166–194 (2007), arXiv:quant-ph/0405098.